HEKAT Categorical Algebra: Master Index

Welcome to the Categorical Algebra Curriculum

This is a **complete, progressive curriculum** for mastering categorical algebra, **working backwards from Kan Extensions** as the ultimate unifying concept.

What You'll Find Here

21 research documents organized in 5 sections: - **7 Core Topics** (comprehensive research) - **10 Prerequisites** (foundational concepts) - **2 Analysis Documents** (structure + applications) - **3 Capstone Documents** (curriculum, Kan extensions mastery)

All designed to work **backwards**: Master prerequisites → Learn core topics → Unify through Kan extensions

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📚 Document Catalog

CORE TOPICS (7 comprehensive research documents)

1. **MONOIDAL-CATEGORIES-COHERENCE.md** (5.5KB)
    - Tensor products and coherence theorems
    - Mac Lane's coherence theorem
    - Braided and symmetric categories
    - String diagrams and TQFT applications
2. **ADJOINT-FUNCTORS-RESEARCH.md** (5.7KB)
    - Free-forgetful adjunctions
    - Universal properties and adjoints
    - RAPL theorem (right adjoints preserve limits)
    - Monadic functors
3. **MONADS-ALGEBRAIC-THEORIES-OPERADS.md** (6.9KB)
    - Monad definition and coherence laws
    - Kleisli and Eilenberg-Moore categories
    - Lawvere theories and operads
    - Programming language effects
4. **ENRICHED-CATEGORY-THEORY-RESEARCH.md** (12KB)
    - V-enriched categories and examples
    - Lawvere metric spaces
    - Weighted limits and Yoneda lemma
    - 17 detailed examples
5. **HIGHER-CATEGORY-THEORY-RESEARCH.md** (11KB)
    - 2-categories and bicategories
    - Quasi-categories and ∞-categories
    - Stable ∞-categories
    - Homotopy type theory connection
6. **TOPOS-THEORY-CATEGORICAL-LOGIC.md** (10KB)
    - Elementary topoi axioms
    - Grothendieck topoi and sheaves
    - Internal logic and Kripke-Joyal semantics
    - Topos as foundations
7. **CATEGORICAL-ALGEBRA-HOPF-TANNAKA.md** (13KB)
    - Hopf algebras (25+ examples)
    - Tannaka-Krein duality
    - Yang-Baxter equation and quantum groups
    - Topological quantum field theory
    - String diagrams and categorical quantum mechanics

PREREQUISITES (10 foundational documents)

Located in `prerequisites/` folder:

1. **01-CATEGORIES-FOUNDATIONAL.md** - Category axioms, 11 examples

2. **02-MORPHISMS-DIAGRAMS.md** - Morphism types, commutative diagrams
3. **03-FUNCTORS.md** - Structure-preserving maps, 12 examples
4. **04-NATURAL-TRANSFORMATIONS.md** - Morphisms between functors, 15+ examples
5. **05-HOM-FUNCTORS-REPRESENTABLE.md** - Representable functors, 10 examples
6. **06-LIMITS-COLIMITS.md** - Universal cones, 15 examples
7. **07-UNIVERSAL-PROPERTIES.md** - Characterizing structures, 15 examples
8. **08-FUNCTOR-CATEGORIES.md** - Categories of functors, Yoneda embedding
9. **09-YONEDA-LEMMA.md** - Central theorem of category theory
10. **10-DUALITY.md** - Opposite categories, Stone duality

ANALYSIS DOCUMENTS (2)

1. **STRUCTURAL-ANALYSIS.md** (37KB)
   - Concept dependency graph and learning hierarchy
   - Three major "schools" of category theory (Australian, French, Quantum)
   - Learning pathways (critical/balanced/deep)
   - Integration matrix: how topics connect
2. **PRACTICAL-IMPLICATIONS.md** (skeleton)
   - Applications across mathematics and computer science
   - Algorithms from categorical structures
   - Design patterns from categorical thinking
   - Implementation guidelines

CAPSTONE DOCUMENTS (3)

1. **CURRICULUM-PROGRESSION.md** (skeleton)
   - 4-level learning architecture
   - Week-by-week critical path (12-16 weeks)
   - Assessment checkpoints
   - Milestone validation
2. **KAN-EXTENSIONS-MASTERY-GUIDE.md** (skeleton)
   - Complete Kan extension definition
   - How all prerequisites lead to Kan extensions
   - 15+ worked examples
   - "All Concepts Are Kan Extensions"
3. **INDEX.md** (this file)
   - Master navigation guide
   - Reading orders by goal and pace

🎯 Reading Order by Goal

Goal 1: Understand Category Theory Basics (4-6 weeks)

Start here if: You want foundational category theory knowledge

1. Prerequisites 01-05: Categories through Hom-functors
2. STRUCTURAL-ANALYSIS: Understand the landscape
3. Review Core Topics 1-2: Monoidal and Adjunctions

Outcome: Solid understanding of categorical fundamentals

Goal 2: Master Kan Extensions (12-16 weeks - CRITICAL PATH)

Start here if: You want to understand the unifying concept

1. Prerequisites 01-07 (2-3 weeks): Categories through Universal Properties
2. Core Topics 1-3 (3-4 weeks): Monoidal, Adjunctions, Monads
3. Prerequisites 08-10 (1-2 weeks): Functor Categories, Yoneda, Duality
4. CURRICULUM-PROGRESSION (1 week): Learning architecture
5. Core Topics 4-7 (3-4 weeks): Enriched, Higher, Topos, Categorical Algebra
6. KAN-EXTENSIONS-MASTERY-GUIDE (2-3 weeks): Synthesis

Outcome: Complete mastery of Kan extensions and how everything unifies

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Goal 3: Explore Applications (8-12 weeks)

Start here if: You want practical applications

1. Prerequisites 01-06 (2 weeks): Basics through Limits
2. MONADS paper (1 week): Programming applications
3. ENRICHED paper (2 weeks): Machine learning, metrics
4. CATEGORICAL-ALGEBRA paper (2 weeks): Physics and quantum computing
5. PRACTICAL-IMPLICATIONS (1-2 weeks): Your domain

Outcome: Ready to apply categorical thinking to your field

Goal 4: Deep Researcher Path (30-36+ weeks)

Start here if: You want comprehensive mastery for research

1. All Prerequisites: Deep study with exercises (8-10 weeks)
2. All Core Topics: Comprehensive study (10-12 weeks)
3. STRUCTURAL-ANALYSIS: Understand landscape deeply (2 weeks)
4. CURRICULUM-PROGRESSION: Full learning architecture (2 weeks)
5. KAN-EXTENSIONS-MASTERY-GUIDE: Complete synthesis (3-4 weeks)
6. External papers and books for each topic

Outcome: Research-ready expertise in categorical algebra

🗺️ Prerequisite Dependency Chain

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Categories (01)
    ↓
Morphisms (02)
    ↓
Functors (03)
    ↓
Natural Transformations (04)
    ↓
Hom-Functors (05)  ←→  Limits/Colimits (06)
    ↓                        ↓
Universal Properties (07)    ↓
    ↓                        ↓
```

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Functor Categories (08) ←──┘
     ↓
Yoneda Lemma (09)
     ↓
Duality (10)
```

Linear path: 01 → 02 → 03 → 04 → 05 → 08 → 09 → 10

Parallel track: 06 → 07 (can learn simultaneously with 05-08)

🎓 Recommended Learning Strategies

For Self-Study

- Read sequentially by prerequisite chain
- Work through all examples
- Take notes on key insights
- Practice with exercises

For Teaching

- Use STRUCTURAL-ANALYSIS to understand landscape
- Adjust pacing based on student background
- Supplement with external books
- Emphasize applications to student interests

For Research

- Deep dive on core topics 4-7
- Master Kan extensions completely
- Study external papers in your specialization
- Connect to your research questions

📖 External Resources by Topic

Category Theory Foundations

- Mac Lane, "Categories for the Working Mathematician"
- Awodey, "Category Theory"
- Leinster, "Basic Category Theory"

Specific Topics

- **Monads**: Moggi (1989), Haskell libraries
- **Enriched**: Kelly, "Basic Concepts of Enriched Categories"
- **Higher**: Lurie, "Higher Topos Theory"
- **Topos**: Mac Lane & Moerdijk, "Sheaves in Geometry and Logic"
- **Hopf/Quantum**: Kassel, "Quantum Groups"

✅ Self-Assessment Checkpoints

Checkpoint 1: After Prerequisites 01-05

- [ ] Can define category axioms
- [ ] Understand morphism types
- [ ] Know 5+ functor examples
- [ ] Can state naturality condition
- [ ] Recognize representable functors

Checkpoint 2: After Prerequisites 06-10

- [ ] Understand limits via universal property
- [ ] Know colimits dually
- [ ] Can state Yoneda lemma
- [ ] Understand opposite category uses
- [ ] Recognize dualities

Checkpoint 3: After Core Topics 1-3

- [ ] Know monoidal structure
- [ ] Understand free-forgetful adjunction
- [ ] Can compute with monads
- [ ] Know Lawvere theories

Checkpoint 4: After Core Topics 4-7

- [ ] Master enriched categories
- [ ] Understand 2-categories
- [ ] Know topos axioms
- [ ] Understand Hopf algebras
- [ ] Ready for Kan extensions

🎯 Success Criteria

You've mastered categorical algebra when you can:

1. **Explain why**: Not just state definitions, understand motivation
2. **Recognize patterns**: See categorical structures in your field
3. **Make connections**: How do topics relate? What unifies them?
4. **Apply insights**: Use categorical thinking to solve problems
5. **Teach others**: Explain concepts clearly to colleagues

📋 Quick Reference: Files by Purpose

For Theory

- Core Topics 1-7
- Prerequisites 01-10

For Applications

- CATEGORICAL-ALGEBRA (quantum, physics)
- MONADS (programming)
- ENRICHED (machine learning)
- PRACTICAL-IMPLICATIONS

For Understanding Structure

- STRUCTURAL-ANALYSIS
- CURRICULUM-PROGRESSION
- PREREQUISITE-TREE

For Mastery

- KAN-EXTENSIONS-MASTERY-GUIDE

For Navigation

- INDEX.md (this file)

🚀 Getting Started

1. **Determine your goal** (basics, mastery, applications, research)
2. **Choose reading order** from section above
3. **Set timeline** (4 weeks to 1 year depending on goal)
4. **Start with Prerequisites** (unless you have background)
5. **Work through systematically**
6. **Use STRUCTURAL-ANALYSIS** to understand connections
7. **End with KAN-EXTENSIONS-MASTERY** for synthesis

📞 Contact & Feedback

This curriculum was generated through deep research and multi-agent synthesis.

Questions? Check the relevant core topic paper or prerequisite document.

Last Updated: 2025-10-27 **Total Documents**: 21 **Total Content**: ~150KB of comprehensive research **Target Audience**: Students, researchers, practitioners applying category theory

Start with the reading order that matches your goal. Welcome to categorical algebra!